

ECON 154: The Global Economy: Trade, Finance, and Policy

Lecture 5 Notes

Markets in Action

How Prices Coordinate Buyers and Sellers

Professor Daniel Riera-Crichton
Bates College

Fall 2026

Contents

1	Introduction: How Do Markets Coordinate Millions of Decisions?	2
2	Markets as Coordination Mechanisms	2
3	Returning to Our Coffee Shop	3
4	The Coordination Problem	4
5	Market Equilibrium	4
6	A Numerical View of Market Equilibrium	5
7	How a Market Shock Appears in the Equations	6
8	What Equilibrium Does—and Does Not—Mean	8
9	Why Prices Matter	9
10	When Markets Are Out of Balance	9

11 When Prices Are Too Low: Shortages	10
12 How Markets Respond to Shortages	10
13 When Prices Are Too High: Surpluses	11
14 How Markets Respond to Surpluses	11
15 The Logic of Price Adjustment	12
16 Visualizing Price Adjustment	13
17 Comparative Statics: What Happens When Markets Change?	15
18 A Four-Step Framework	15
19 Example 1: Increased Demand for Coffee	15
20 Example 2: Lower Production Costs	17
21 Example 3: A Negative Supply Shock	18
22 Reading Economic News	19

Learning Objectives

After completing this lecture you should be able to

1. Explain how markets coordinate the decisions of consumers and firms.
2. Define market equilibrium.
3. Explain why shortages and surpluses create pressure for prices to change.
4. Use supply and demand to determine equilibrium price and quantity.
5. Distinguish movements along curves from shifts in curves.
6. Define consumer surplus and producer surplus.
7. Explain why voluntary exchange creates gains from trade.

1 Introduction: How Do Markets Coordinate Millions of Decisions?

Over the past two lectures we have studied the two sides of every market separately. Lecture 3 focused on consumers. Consumers decide whether to purchase a good or service by comparing the benefit they expect to receive with the price they must pay. Lecture 4 focused on firms. Firms decide whether to produce a good or service by comparing the revenue they expect to receive with the cost of production. At this point we understand each side of the market independently. Consumers have plans. Firms have plans. A natural question now emerges.

How do these independent plans become a market outcome?

Suppose consumers would like to purchase 220 cups of coffee this morning. Coffee shops, however, only plan to prepare 150 cups. Who decides what happens? Who determines the price? Who determines how many cups are actually sold? This lecture answers these questions.

2 Markets as Coordination Mechanisms

One of the central insights of economics is that markets coordinate the decisions of millions of individuals without requiring a central planner. Every consumer makes decisions independently. Every producer makes decisions independently. Yet, most of the time, markets function remarkably smoothly. Coffee shops prepare approximately the number of coffees customers wish to buy.

Grocery stores stock approximately the amount of milk consumers demand. Airlines schedule approximately the number of flights travelers wish to take. This coordination occurs largely through prices.

Definition

A market is a mechanism through which buyers and sellers interact to exchange goods and services. Prices communicate information to both consumers and producers and help coordinate their decisions.

Notice that markets do not require buyers and sellers to know one another personally. A consumer purchasing coffee does not need to know who harvested the beans, transported them across the ocean, roasted them, or delivered them to the café. Likewise, the coffee producer does not need to know who will ultimately purchase today's latte. The market price connects these decentralized decisions.

3 Returning to Our Coffee Shop

Throughout Lectures 3 and 4 we followed a neighborhood coffee shop. Now imagine the following situation. At a price of \$4 per latte, customers wish to purchase

220

lattes. Coffee shops, however, are willing to prepare only

150

lattes. Immediately a problem appears. Consumers want to purchase more coffee than firms are willing to provide. Economists call this situation a shortage. Now imagine instead that the market price is \$8. Consumers now wish to purchase only

90

lattes. Coffee shops, however, have prepared

180

lattes. Now firms wish to sell more coffee than consumers are willing to buy. Economists call this situation a surplus. Neither situation can persist indefinitely. Markets contain forces that tend to move prices toward a point where planned purchases and planned sales coincide.

4 The Coordination Problem

Figure 1 illustrates the challenge.

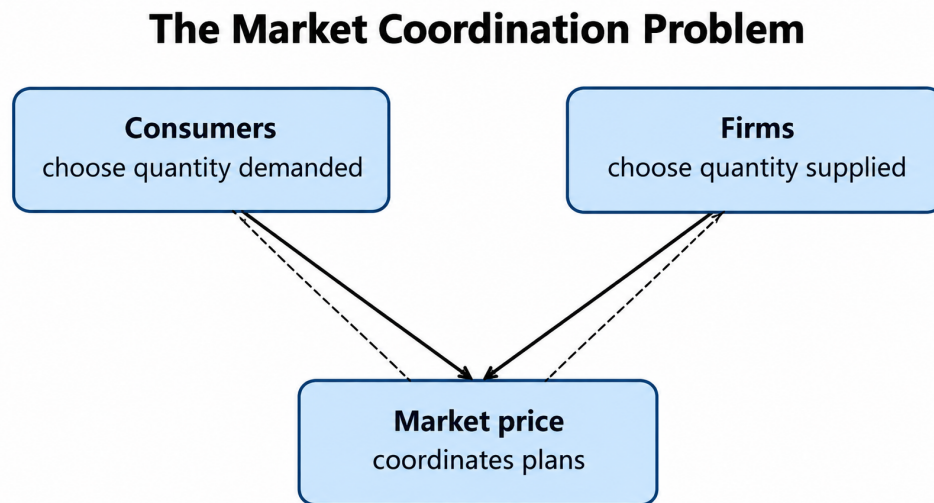


Figure 1: Consumers and firms make independent decisions. Prices coordinate these plans.

Consumers answer one question.

How much do I want to buy?

Firms answer another.

How much do I want to sell?

Markets answer a third.

What price allows both plans to be simultaneously satisfied?

5 Market Equilibrium

Economists call the answer to this coordination problem *market equilibrium*.

Definition

Market equilibrium is the situation in which the quantity consumers wish to purchase equals the quantity firms wish to sell. At the equilibrium price,

$$Q_D = Q_S.$$

The word “equilibrium” sometimes creates confusion. In everyday language equilibrium often means “perfect” or “ideal.” In economics it means neither. Equilibrium simply refers to a situation in which buyers and sellers have no immediate reason to change the market price.

6 A Numerical View of Market Equilibrium

The graphical analysis is the most important way to build intuition about equilibrium. But the same logic can also be represented algebraically. A common introductory formulation writes quantity demanded and quantity supplied as linear functions of price:

$$Q_D = a - bP, \quad b > 0,$$

and

$$Q_S = c + dP, \quad d > 0.$$

The negative coefficient on price in the demand equation captures the law of demand: holding other determinants of demand constant, a higher price reduces quantity demanded. The positive coefficient on price in the supply equation captures the law of supply: holding other determinants of supply constant, a higher price increases quantity supplied.

Economic Intuition

The equations contain exactly the same information as the demand and supply curves. A change in the good’s own price produces a **movement along** a given equation or curve. A change in another determinant of demand or supply changes the **equation itself**, which corresponds to a shift of the curve.

Consider the market for campus lunch bowls:

$$Q_D = 140 - 4P,$$

$$Q_S = 20 + 2P.$$

At equilibrium,

$$Q_D = Q_S.$$

Therefore,

$$140 - 4P = 20 + 2P.$$

Solving,

$$120 = 6P,$$

so

$$P^* = 20.$$

Substituting the equilibrium price into either equation,

$$Q^* = 140 - 4(20) = 60.$$

Thus the market equilibrium is

$P^* = 20, \quad Q^* = 60.$

The algebra does not introduce a new economic idea. It simply gives us another way to locate the intersection of demand and supply.

7 How a Market Shock Appears in the Equations

Suppose a new dining trend makes campus lunch bowls more popular. At every possible price, consumers now want to buy 24 more bowls. The original demand equation was

$$Q_D = 140 - 4P.$$

The new demand equation becomes

$$Q'_D = 164 - 4P.$$

Supply is unchanged:

$$Q_S = 20 + 2P.$$

The new equilibrium satisfies

$$164 - 4P = 20 + 2P.$$

Therefore,

$$144 = 6P, \quad P_1^* = 24.$$

The new equilibrium quantity is

$$Q_1^* = 20 + 2(24) = 68.$$

So the positive demand shock produces

$$P^* : 20 \rightarrow 24, \quad Q^* : 60 \rightarrow 68.$$

This is precisely what the graph predicts: a rightward shift of demand raises both equilibrium price and equilibrium quantity. But the equations also help us see the **adjustment process**. Immediately after demand shifts, suppose price is still at the old equilibrium price, $P = 20$. Then

$$Q'_D = 164 - 4(20) = 84,$$

while

$$Q_S = 20 + 2(20) = 60.$$

There is a shortage of

$$84 - 60 = 24.$$

That shortage puts upward pressure on price. As price rises from 20 toward 24, consumers move **along the new demand curve**, so quantity demanded falls. At the same time, firms move **along the existing supply curve**, so quantity supplied rises. At $P = 24$,

$$Q'_D = Q_S = 68,$$

and the market reaches its new equilibrium.

Economic Intuition

A market shock and the subsequent price adjustment are different steps.

First: a non-price determinant changes, shifting demand or supply.

Then: at the old price, the market has a shortage or surplus.

Finally: price adjusts, causing movements along the relevant curves until $Q_D = Q_S$ again.

Real-World Example

Try this in class.

Suppose instead that production technology improves and supply increases by 18 units at every price.

1. How would the supply equation change?
2. At the old equilibrium price, would there be a shortage or a surplus?
3. In which direction would price adjust?
4. As price changes, how would buyers and sellers move along their curves?
5. What would happen to the new equilibrium price and quantity?

The goal is not complicated algebra. The goal is to connect the equation, the graph, and the economic adjustment process.

8 What Equilibrium Does—and Does Not—Mean

Students often misunderstand the meaning of equilibrium. Equilibrium does **not** imply that everyone is happy. Some consumers would prefer lower prices. Some producers would prefer higher prices. Equilibrium does not imply that income is equally distributed. Nor does it imply that the market outcome is necessarily socially optimal in every circumstance. Rather, equilibrium means only that the quantity buyers plan to purchase equals the quantity sellers plan to provide.

Common Mistake

Equilibrium does not mean fairness. It means balance between planned purchases and planned sales.

9 Why Prices Matter

Prices play two distinct roles. First, prices ration scarce goods. When prices rise, some consumers purchase less. Second, prices provide incentives for firms. Higher prices encourage additional production because more units become profitable. Thus, prices simultaneously influence both sides of the market. This dual role explains why economists often describe prices as signals.

Economic Intuition

Prices transmit information. Consumers observe prices and decide how much to buy. Producers observe the same prices and decide how much to produce. Without requiring direct communication, prices coordinate millions of independent decisions.

Ask an Economist

Student

“Why doesn’t the government simply tell every coffee shop how many lattes to make?”

Economist

Because the necessary information is dispersed across millions of consumers and producers. Every consumer knows his or her own preferences. Every producer knows his or her own costs. Prices summarize this information and allow decentralized decision-making without requiring a central planner to know everyone’s preferences and production possibilities.

Key Takeaways

1. Consumers and firms make decisions independently.
2. Markets coordinate these decisions through prices.
3. Market equilibrium occurs when quantity demanded equals quantity supplied.
4. Equilibrium does not imply fairness or perfection.
5. Prices simultaneously influence consumers and producers.
6. The next section studies what happens when prices are above or below equilibrium.

10 When Markets Are Out of Balance

In the previous section we introduced the concept of market equilibrium. At the equilibrium price, the quantity consumers wish to purchase equals the quantity firms wish to sell. But what happens if the market price is not at its equilibrium level? Suppose a price is somehow established that is

either too low or too high. Will the market simply remain there? The answer is almost always no. Markets contain forces that tend to push prices toward equilibrium. Understanding these forces is one of the central goals of introductory economics.

11 When Prices Are Too Low: Shortages

Imagine that every coffee shop in town decides to sell lattes for

\$4.

At this price, many consumers decide to purchase coffee. Some buy their usual morning drink. Others who might have stayed home now decide that coffee is inexpensive enough to justify the purchase. Demand increases. Firms, however, face a different calculation. At \$4, some additional lattes are simply not profitable to produce. Coffee shops therefore prepare fewer drinks than consumers wish to purchase. Suppose consumers wish to purchase 220 lattes, while coffee shops prepare only 150. Immediately, 70 customers cannot obtain the coffee they would like to buy. Economists call this situation a shortage.

Definition

A shortage occurs when

$$Q_D > Q_S.$$

Consumers wish to purchase more than firms are willing to provide at the current price.

Notice that the shortage does not arise because coffee has become physically scarce. The shortage arises because the market price is too low to balance planned purchases and planned sales.

12 How Markets Respond to Shortages

What happens when consumers compete for a product that is in short supply? Several outcomes become possible. Customers may

- wait in line,
- arrive earlier,
- search other stores,
- offer to pay higher prices,

- substitute toward other products.

Producers also observe the situation. They notice that coffee sells immediately. Customers continue arriving. Some leave disappointed. The existing price no longer reflects market conditions. Higher prices now become attractive. Why? Because higher prices accomplish two things simultaneously. First, some consumers decide not to purchase coffee. Second, coffee shops find it profitable to prepare additional drinks. The shortage therefore begins to shrink.

Economic Intuition

A price increase affects both sides of the market. Consumers purchase less. Producers supply more. These two responses move the market toward equilibrium.

13 When Prices Are Too High: Surpluses

Now consider the opposite situation. Suppose coffee shops decide to charge

\$8

for every latte. At this higher price, many consumers decide that coffee is no longer worth purchasing. Demand falls. Coffee shops, however, have already prepared a large quantity because the high price makes production profitable. Suppose firms prepare 180 lattes, while consumers purchase only 90. Ninety drinks remain unsold. Economists call this situation a surplus.

Definition

A surplus occurs when

$$Q_S > Q_D.$$

Firms wish to sell more than consumers wish to purchase at the current price.

14 How Markets Respond to Surpluses

Unsold inventory creates pressure on producers. Coffee that remains unsold generates little or no revenue. Firms therefore have incentives to reduce prices. Lower prices encourage additional consumers to purchase coffee. At the same time, lower prices reduce the profitability of producing very large quantities. Some firms reduce production. The surplus gradually disappears. Again, notice that one price change influences both consumers and producers simultaneously.

15 The Logic of Price Adjustment

The adjustment process can now be summarized.

Price Too Low



Shortage



Upward pressure on prices



Demand decreases Supply increases



Equilibrium

Price Too High



Surplus



Downward pressure on prices

$\Downarrow$

Demand increases Supply decreases

 $\Downarrow$

Equilibrium

16 Visualizing Price Adjustment

Only now are we ready to summarize the adjustment process graphically. Figure 2 combines the demand and supply curves introduced in the previous two lectures.

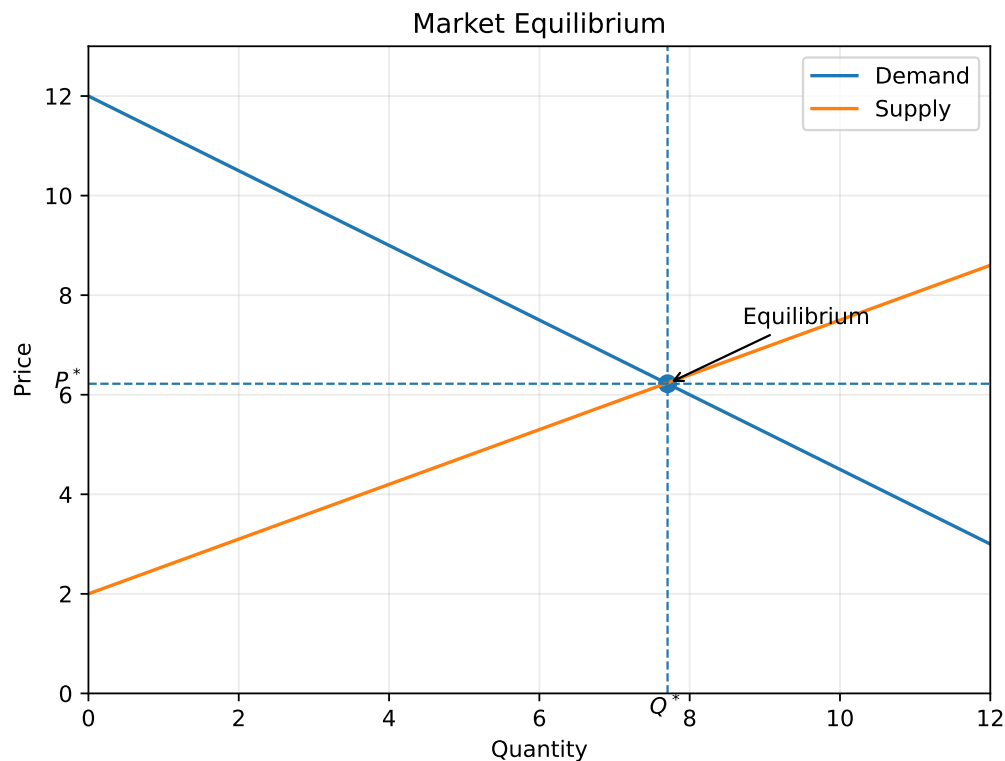


Figure 2: Supply and demand jointly determine the equilibrium price and quantity.

The graph does not replace the reasoning developed above. Instead, it summarizes the adjustment process in a single picture.

Common Mistake

Students often memorize “shortage means price goes up.” The deeper explanation is more important. Prices rise because shortages create incentives for consumers and producers to change their behavior.

Market Watch**Market Watch**

During the COVID-19 pandemic, many grocery stores temporarily experienced shortages of products such as toilet paper, disinfectants, and flour. Were these shortages caused by producers suddenly forgetting how to manufacture these goods? Or did rapid increases in demand temporarily exceed available supply? Think about this question before reading the next section.

Ask an Economist**Student**

“If markets naturally move toward equilibrium, why do shortages sometimes persist?”

Economist

Because prices do not always adjust immediately. Contracts, government regulations, menu costs, capacity constraints, and uncertainty can all slow the adjustment process. These frictions become especially important in labor markets, housing markets, and international markets, which we will study later in the semester.

Checkpoint

At this stage we understand

1. why shortages occur,
2. why surpluses occur,
3. why prices tend to adjust,
4. how prices coordinate consumers and producers.

One important question remains. How do economists use supply and demand to predict what happens when demand or supply changes? The next section introduces comparative statics and applies the equilibrium model to real-world events.

17 Comparative Statics: What Happens When Markets Change?

So far we have assumed that consumers' preferences and firms' production conditions remain unchanged. Under those circumstances, prices naturally move toward equilibrium. Real economies, however, rarely remain unchanged. Consumers' tastes evolve. Technology improves. Taxes change. Weather affects agricultural production. International events disrupt supply chains. Each of these events changes market conditions. Economists refer to the analysis of one change at a time as *comparative statics*.

Definition

Comparative statics studies how a market equilibrium changes after one underlying determinant of demand or supply changes, holding all other factors constant.

Rather than memorizing outcomes, economists follow a systematic reasoning process.

18 A Four-Step Framework

Whenever you encounter an economic news story, ask four questions.

1. What changed?
2. Does the change affect demand or supply?
3. Which curve shifts?
4. How does the new equilibrium price and quantity compare with the old equilibrium?

This framework will become one of the most important analytical tools used throughout this course.

19 Example 1: Increased Demand for Coffee

Suppose a new medical study concludes that moderate coffee consumption reduces the risk of several diseases. Consumers now value coffee more highly.

Step 1

What changed? Consumer preferences.

Step 2

Which side of the market changes? Demand.

Step 3

Which curve shifts? The demand curve shifts to the right. Supply does not immediately change.

Step 4

What happens? The new equilibrium has

$$P \uparrow$$

and

$$Q \uparrow$$

because consumers wish to purchase more coffee at every possible price.

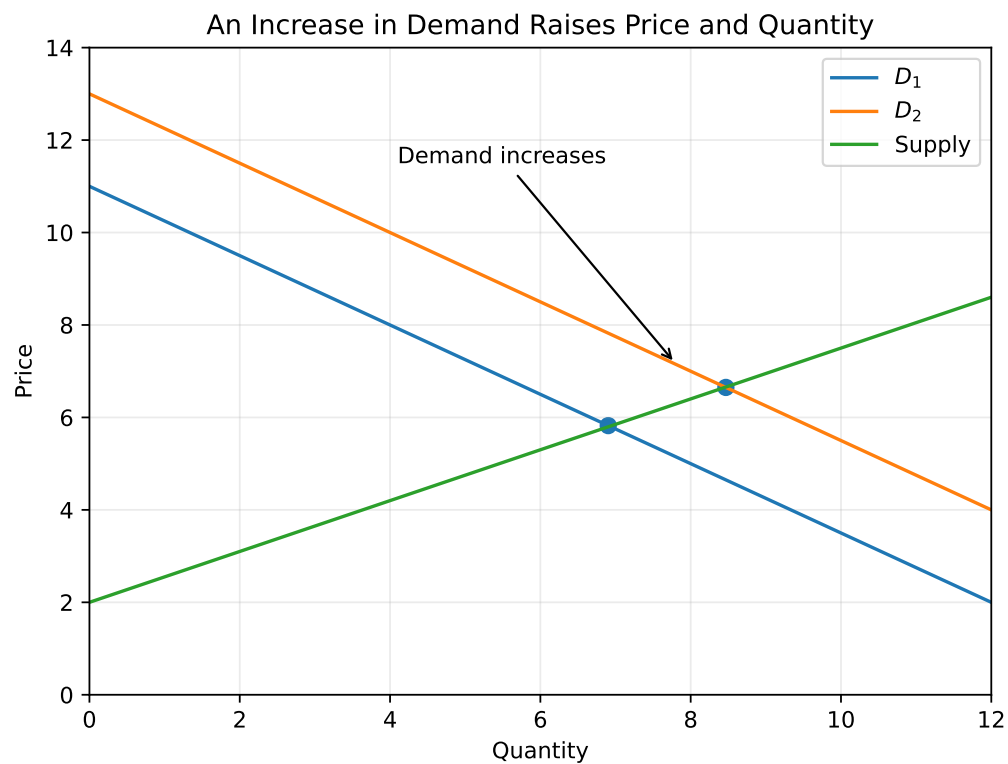


Figure 3: An increase in demand raises both equilibrium price and equilibrium quantity.

20 Example 2: Lower Production Costs

Now suppose a technological improvement allows coffee shops to prepare drinks more efficiently. The cost of producing each latte falls.

Step 1

What changed? Technology.

Step 2

Which side changes? Supply.

Step 3

Which curve shifts? Supply shifts to the right. Demand does not change.

Step 4

The new equilibrium has

$$P \downarrow$$

and

$$Q \uparrow$$

Consumers benefit because coffee becomes less expensive. Firms produce more because production has become cheaper.

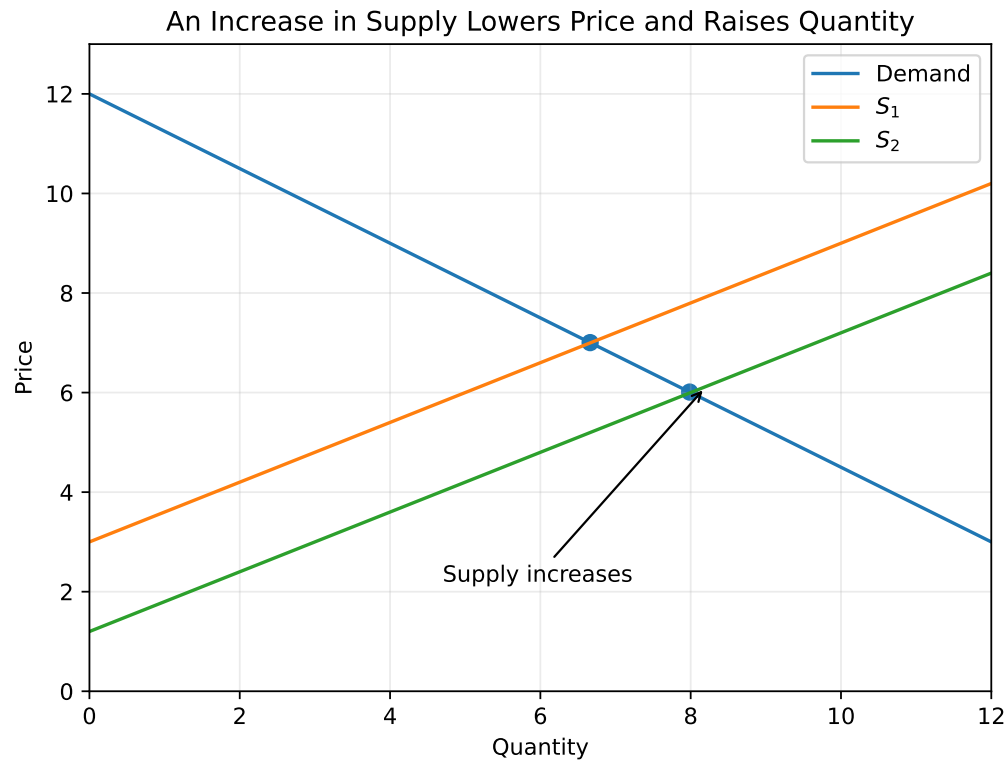


Figure 4: An increase in supply lowers equilibrium price while increasing equilibrium quantity.

21 Example 3: A Negative Supply Shock

Consider a severe frost in Brazil that damages coffee crops. Coffee beans become more difficult to obtain. Production costs rise. Using the four-step framework,

1. Weather changed.
2. Supply changes.
3. Supply shifts left.
4. Equilibrium price rises while equilibrium quantity falls.

Notice that this conclusion follows directly from our framework. There is no need to memorize another rule.

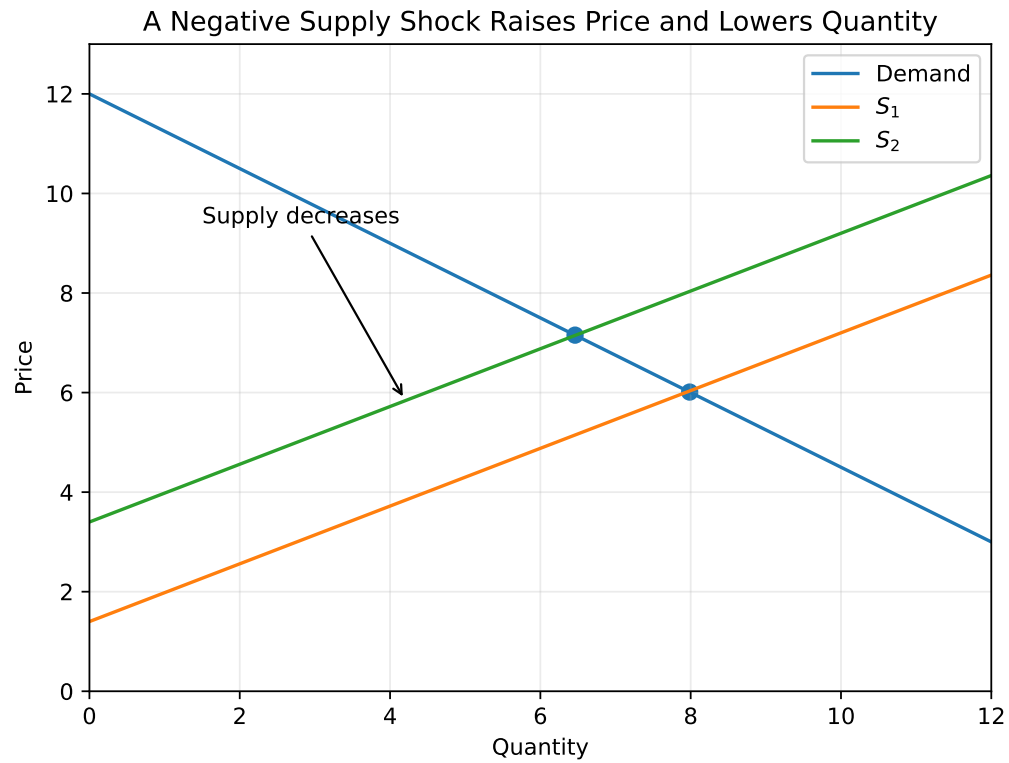


Figure 5: A negative supply shock raises prices and reduces equilibrium quantity.

22 Reading Economic News

Most economic news stories can be interpreted using the same sequence of questions.

Table 1: Applying the Four-Step Framework

News Event	Which Curve?	Likely Result
Coffee becomes fashionable	Demand	Higher price, higher quantity
Improved coffee machines	Supply	Lower price, higher quantity
Brazilian drought	Supply	Higher price, lower quantity
Increase in wages	Supply	Higher price, lower quantity
Population growth	Demand	Higher price, higher quantity

Market Watch**Market Watch**

When reading economic news this week, choose one headline involving food, energy, housing, transportation, or technology. Instead of asking whether prices increased or decreased, ask

What changed first?

The answer usually identifies whether demand or supply shifted.

Common Mistake

Students often begin with the observed price. Economists begin with the underlying shock. Prices are outcomes. Demand and supply shifts are causes.

Ask an Economist**Student**

“If both price and quantity increase, can I conclude that demand increased?”

Economist

Not necessarily. That conclusion is valid only if supply remains unchanged. If both demand and supply change simultaneously, additional information is needed. Beginning in the next lecture, we will analyze multiple shocks occurring at the same time.

Checkpoint

You can now analyze a market systematically. Whenever conditions change,

1. Identify the shock.
2. Decide whether demand or supply changes.
3. Determine which curve shifts.
4. Predict the new equilibrium.

This framework will be used repeatedly throughout the remainder of the course, from labor markets and financial markets to tariffs, international trade, exchange rates, migration, and climate-related economic shocks.